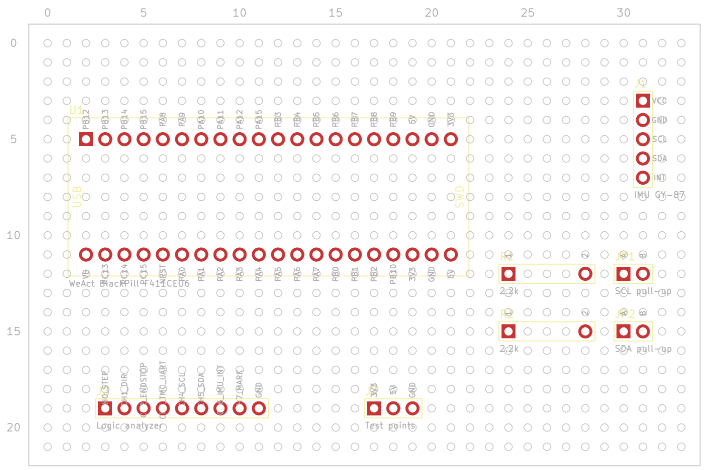
0
5
10
15
20
25
30
0
5
10
15
20
U1
PB12
PB13
PB14
PB15
PA8
PA9
PA10
PA11
PA12
PA15
PB3
PB4
PB5
PB6
PB7
PB8
PB9
5V
GND
3V3
USB
SWD
VB
PC13
PC14
PC15
NRST
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PB0
PB1
PB2
PB10
3V3
GND
5V
WeAct BlackPill F411CEU6
J1
VCC
GND
SCL
SDA
INT
IMU GY-87
R1
1
2
2.2k
JP1
A
B
SCL pull-up
R2
1
2
2.2k
JP2
A
B
SDA pull-up
J2
CH0_STEP
CH1_DIR
CH2_ENDSTOP
CH3_TMC_UART
CH4_SCL
CH5_SDA
CH6_IMU_INT
CH7_MARK
GND
Logic analyzer
J3
3V3
5V
GND
Test points